



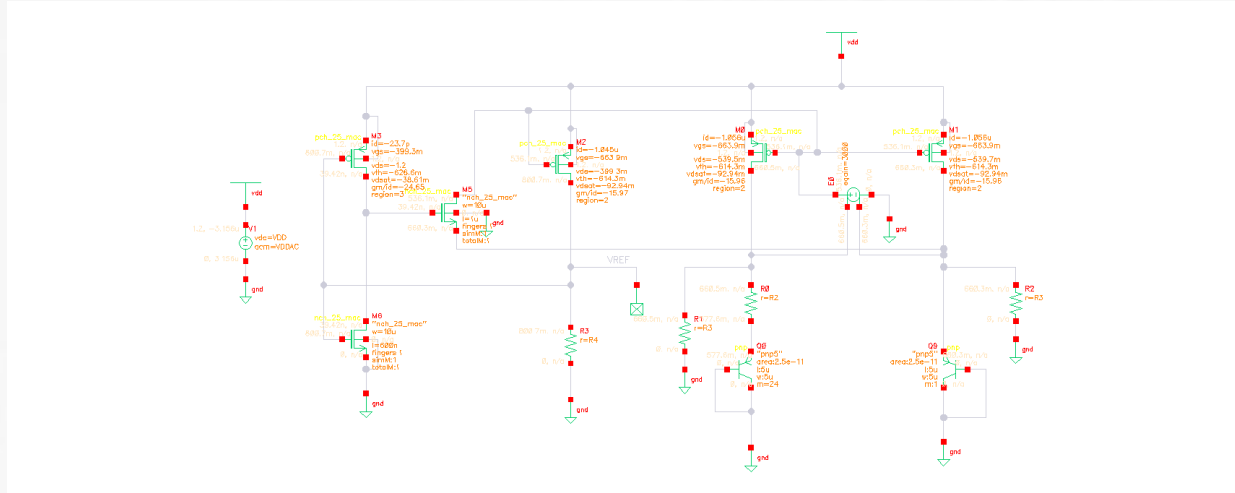
Analog IC Design

CMOS Technology

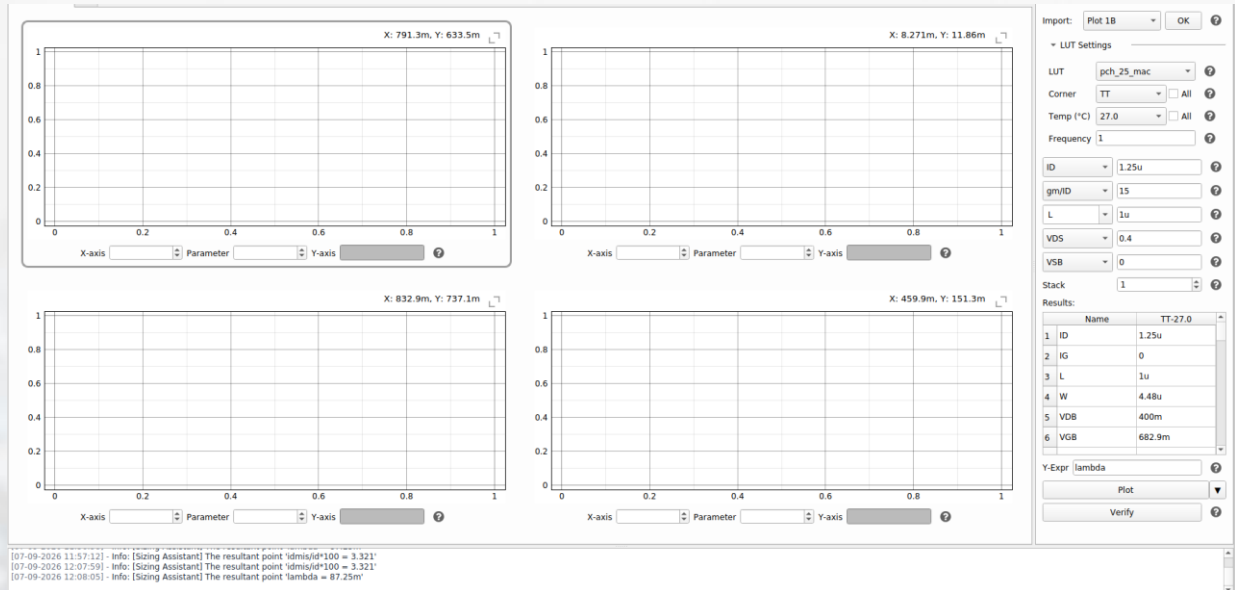
MOHAMMED NASR EL-DIN

BGR Design

Ideal error amplifier



Let $I = 1.25\mu A$, $L = 1\mu m$, $\frac{gm}{I_D} = 15$



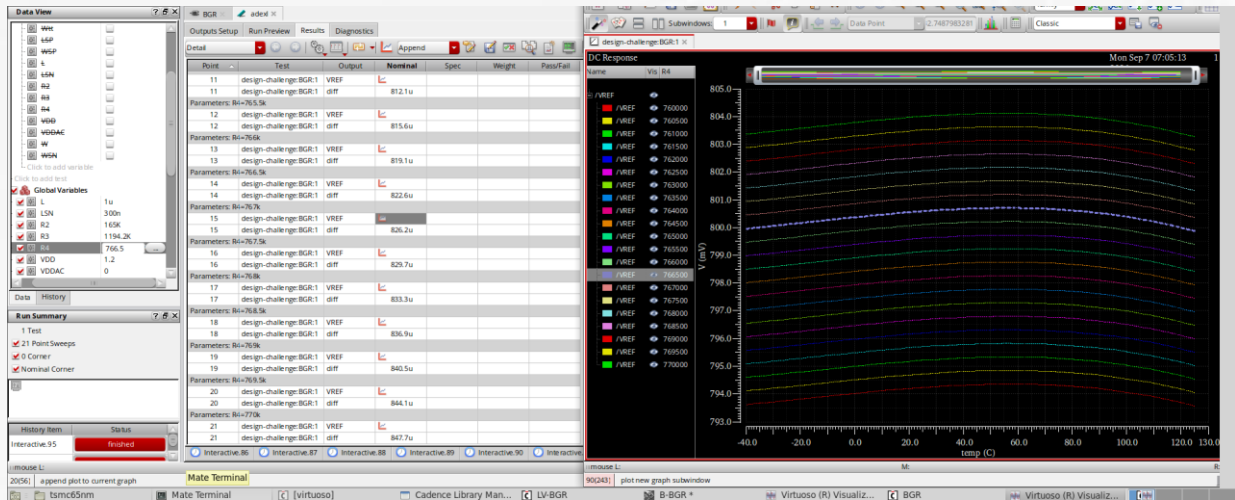
$$W = 5\mu m$$

For BJTs, take $n = 24 \Rightarrow \Delta V_{BE} = 82.6mV$

Note, start-up circuit same as the mentioned in the file.

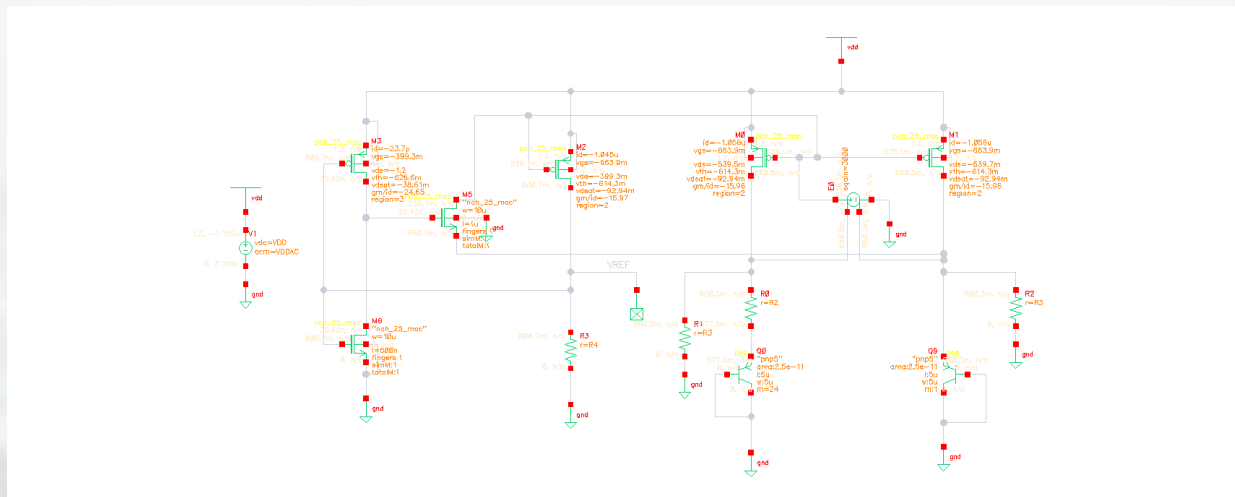
Let the current that will pass through R_2 is $0.6\mu A \Rightarrow R_2 \approx 165K$

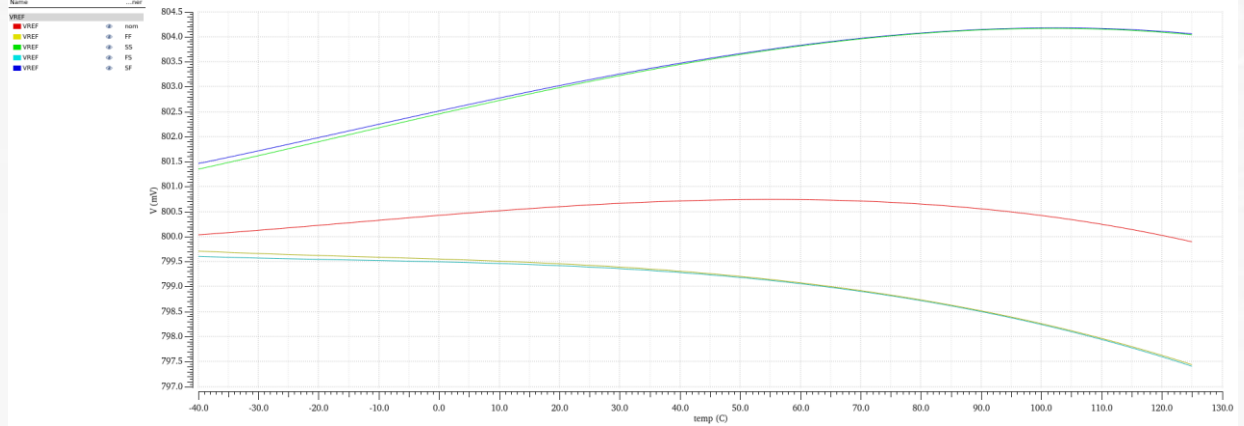
Let $V_{BE} \approx 0.65V \Rightarrow R_3 = \frac{0.65}{0.55\mu A} \approx 1200K$, tuning $R_3 = 1194.2K$



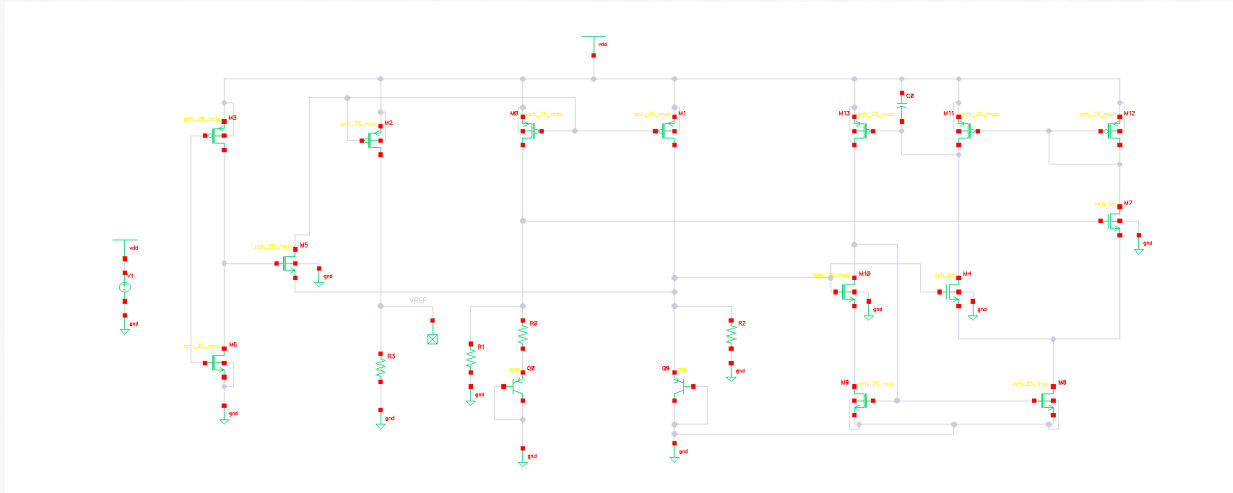
Sweep on $R_4 \Rightarrow R_4 \approx 766.5K$

DC OP





Actual error amplifier



Taking the same values from the file and sweeping the parameters to get the best results

NMOS input: $W/L = 30/20$

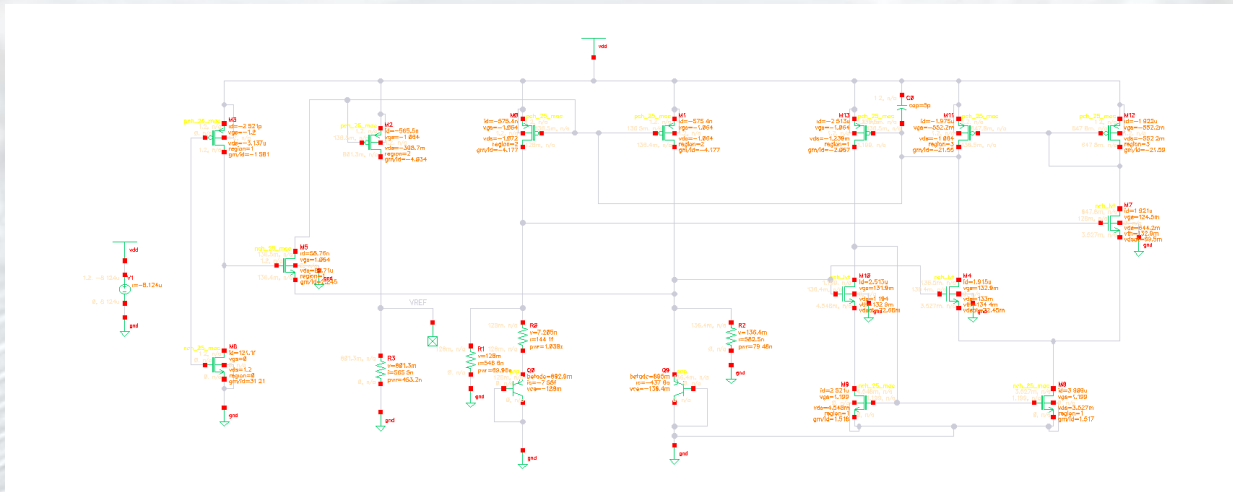
PMOS load: $W/L = 80/1$

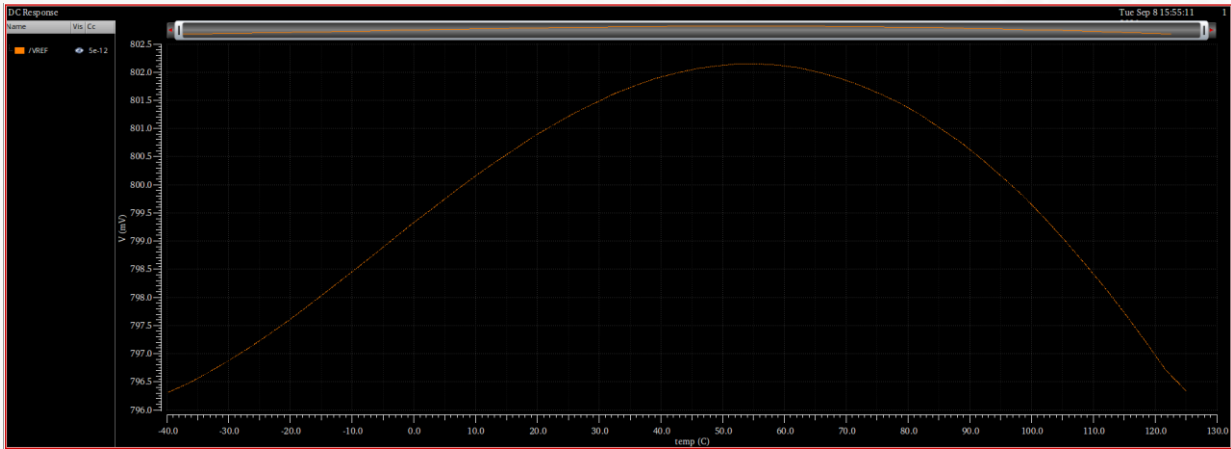
NMOS tail $W/L = 8.4/1.97$

Same sizing for the start-up circuit

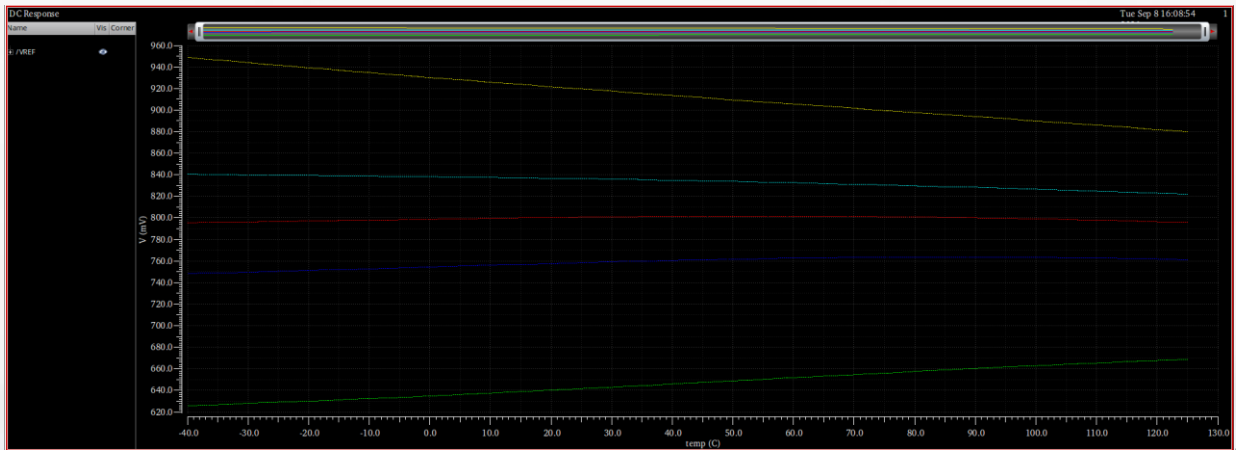
Tuning the BGR PMOS MOSFETs: $W/L = 0.62/5.5$

DC OP





5.8mv variation



Test	Output	Nominal	Spec	Weight	Pass/Fail
design-challenge:BGR-ota:1	VREF				
design-challenge:BGR-ota:1	phase margin	91.35			

Transient

